

Total No. of Questions : 4]

SEAT No. :

PC149

[6360]-158

[Total No. of Pages :1

T.E. (Electronics and Telecommunication) (Honors) (Insem)

PRINCIPLES OF ROBOTICS

(2019 Pattern) (Semester- I) (304181HR)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) *Answer Q.1 or Q.2 and Q.3 or Q.4.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Assume suitable data if required.*

- Q1)** a) Sketch and explain Spherical (Polar) Robot configuration. [5]
b) Explain Robot components/Anatomy with neat sketch. [5]
c) Sketch and explain various joints in Robot. [5]

OR

- Q2)** a) Sketch and explain various DOF's associated with arm body & Robot wrist. [5]
b) Define and explain accuracy, resolution & precision of Robots. [5]
c) Define Industrial Robots. State 3 Asimov's laws of Robotics. [5]

- Q3)** a) Explain selection characteristics of drives. [6]
b) Classify and state advantages of Hydraulic Motors. [4]
c) Write short notes on any 1 :- [5]
i) DC Servo Motor
ii) AC Servo Motor
iii) Stepper Motor
iv) Brushless DC Motor

OR

- Q4)** a) Compare Hydraulic, Pneumatic & electric drive according to their merits & demerits. [6]
b) Sketch and explain Planetary gearbox for Robots. [4]
c) Explain Stepper Motor with neat sketch. State its salient features. [5]

